◉ DOMAIN 01 / AI SYSTEMS & AGENTIC WORKFLOW AUTOMATION

CLINIFLOW.

An end-to-end agentic clinical workflow for SME clinics — powered by an OpenAI-compatible LLM (ILMU GLM 5.1 in production) as the central reasoning core.

Dependency Free

Who am I ?

ShaoXian

Year 3 Software Engineering Student @ UM

Meet Dr Lim who sees 40 patients a day

He spent more time *typing* than talking today.

Doctors spend **30–50%** of every shift on documentation

Patients repeat their history multiple times per visit

Records live across paper, Excel, and disconnected tools

Patients leave without understanding their treatment

THREE FATAL FLAWS

Every tool today stops short of the workflow — and most have no clinical-safety net.

FLAW · 01

Passive Note-Takers

"AI scribes" only transcribe. They don't reason, ask follow-ups, query patient history, or detect drug interactions.

▸ CliniFlow reasons, plans, and orchestrates — not just transcribes.

FLAW · 02

Hospital-Scale Only

Enterprise EMRs cost six figures and need IT teams. SME clinics — the majority of Malaysian primary care — get nothing built for them.

▸ Cloud SaaS · Docker Compose · 1-day deploy · no on-prem hardware.

FLAW · 03

No Patient Memory · No Safety Net

Tools see each visit in isolation. No drug-interaction check, no allergy lookup, no dose validation — clinical risk hits the patient.

▸ Neo4j drug-knowledge graph · Evaluator agent · 6 validators (DDI, allergy, dose, pregnancy, hallucination, completeness).

3 AGENTS · 3 PHASES · 1 LLM CORE · 2 GRAPHS (PATIENT + DRUG)

An *agentic workflow* for the entire visit.

Three specialised agents — Pre-Visit Intake, Report (SOAP + bilingual summary), and Evaluator (clinical-safety gate) — orchestrated by an LLM. ReAct loops, per-agent tool allowlists, stateful across visits. Two graphs: patient knowledge + drug knowledge. Self-correcting via Hermes (style-only).

Agent 1

Intake Agent

Conversational symptom collection before arrival.

- Multi-turn ReAct loop
- Read-only tools · graph KB lookup
- Strict-JSON slot extraction
- Structured pre-visit report

Agent 2

Report Agent

SOAP draft from voice or text · graph drug-interaction check.

- 3 input modes: live · upload · text / Whisper STT / SOAP + ICD-10 / Evaluator gate · 6 validators / Hermes style-only loop

Agent 3

Evaluator Agent

Inspired by the idea of harness engineering, agent evaluate

- drug drug interaction
- drug dosage validation
- report completeness
- report agent hallunication

01 · PRE-VISIT · INTAKE

Patient *talks first.* Doctor reads a brief.

A ReAct-loop agent runs the multi-turn conversation: it queries the Neo4j patient graph (get_patient_context, get_visit_history) for allergies, active meds and prior diagnoses before each question, then asks only what's missing. When the agent decides intake is complete, a separate strict-JSON extract_slots() call produces the structured pre-visit report — on Dr. Lim's screen before the patient sits down.

ReAct Loop | Tool Allowlist · Read-Only | Graph KB · Multi-Hop Query | Strict-JSON Slot Extraction | EN · MVP

CLINIFLOW · INTAKE
Hi! Before your visit with Dr. Lim — what's bothering you today?

PATIENT
I've had a bad cough for 3 days, sometimes a bit of blood.

CLINIFLOW · INTAKE
[graph KB · no penicillin allergy on file]
Cough for 3 days, with some blood. Any fever?

PATIENT
Yes, last night 38.5°C.

CLINIFLOW · INTAKE
Got it — intake complete. Generating your pre-visit report now.

▸ STRUCTURED REPORT GENERATED · DOCTOR NOTIFIED

02 · VISIT · REPORT AGENT

3 input modes. *One pipeline.*

Live recording (MediaRecorder), file upload (≤25 MB), or manual text — all converge on the same Report Agent. Whisper transcribes. The agent generates SOAP, runs a graph multi-hop drug-interaction check, hands off to the Evaluator (6 validators), doctor free up his hand

[S]	Productive cough × 3d, occasional haemoptysis. Fever 38.5°C since yesterday.
[O]	T: 38.4°C · BP 128/82 · SpO_2 96% · chest crackles RLL
[A]	Likely community-acquired pneumonia — RLL. R/O TB.
[P]	CXR · sputum AFB · amoxicillin 500mg TDS × 7d.
⚠ GRAPH	Evaluator: 0 CRITICAL · 1 HIGH (dose at upper bound) · DDI ✓ · Allergy ✓ · Pregnancy n/a · Hallucination ✓

▸ DRAFT PERSISTED · EVALUATOR GATE · AWAITING DOCTOR FINALIZE

Live Record
MEDIARECORDER

File Upload
≤25MB · MP3/WAV

Manual Text
DEFAULT TAB

Whisper-1 STT | SOAP + ICD-10 Extract | Drug Interaction · Graph Multi-Hop | Evaluator Gate · Soft-Block on CRITICAL

03 · POST-VISIT · FINALIZE

Same agent. *Tool call on finalize.*

Phase 3 is not a separate agent. When the doctor finalizes (after acknowledging any CRITICAL Evaluator finding with a written reason), the Report Agent calls a tool that produces the bilingual EN + MS patient summary as well as the e-perscription generation. Follow up reminder will be sent via Whatsapp

Bilingual EN + MS | Same Transaction as Report Write | Red-Flag Symptom Extraction | Patient Portal Push

DIAGNOSIS	You have a chest infection (pneumonia). It's treatable.
MED · 01	Amoxicillin 500mg — 1 capsule, 3× daily, 7 days. With food.
MED · 02	Paracetamol 500mg — if fever >38°C. Max 4× daily.
RED FLAG	Difficulty breathing · chest pain · cough up more blood — go to ER.
FOLLOW-UP	Return in 3 days · or sooner if breathing worsens.

▸ DELIVERED EN + MS · CACHED IN MEDICAL_REPORTS

EDIT-DIFF CAPTURE · GRAPH-STORED RULES · FENCED FROM CLINICAL REASONING

The system learns from doctor edits — but only how to write, never what to write.

ADAPTIVE LOOP · STYLE-ONLY

Edit-Diff → Rule → Prompt

1. Doctor finalizes SOAP
2. Edit-diff extracted (was → now)
3. Hermes proposes a rule — phrasing or structure only
4. Stored in Neo4j with confidence + provenance
5. Injected into next prompt as few-shot examples
6. A/B test: edit-rate delta with vs without rules

HERMES_FENCE · CLINICAL FIREWALL

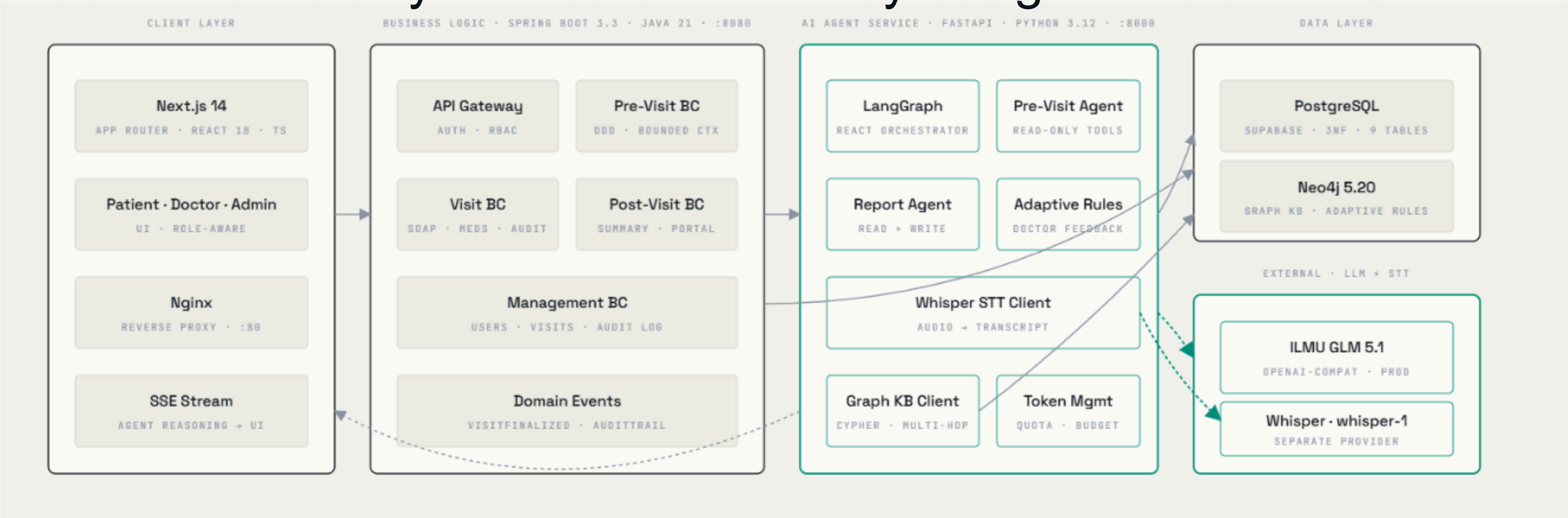
Style only · Never reasoning

```
# ALLOWED — phrasing, abbreviation style, ordering, structure
# BLOCKED — diagnosis, dose, contraindication,
# red-flag thresholds, ICD-10 selection.

# Doctor-in-the-loop guarantee:
# AI draft is always watermarked.
# Explicit Finalize button required.
# PDPA audit row written every time.
```

NEXT.JS · SPRING BOOT · FASTAPI · POSTGRES + NEO4J · SENTRY · UPTIME ROBOT

Frontend talks to Spring Boot only · Spring Boot is the only writer to Postgres · FastAPI is the only LLM caller. PDPA-by-design · 4-role RBAC.



- 3 LangGraph agents · ReAct loops · per-agent tool allowlists
- REST + JDBC + Bolt protocol
- SSE stream · agent reasoning to UI
- VisitFinalized event · audit trail · Sentry across 3 services

SIX STAGES · ONE STATEFUL WORKFLOW · DOCTOR EDITS LEARN

INTAKE → CONSULT → SOAP → *REVIEW* → *FINALIZE* ↺

01
BOOTSTRAP

- Patient logs in
- Visit row created
- agent_turns init
- Audit begins

02
PRE-VISIT

- Intake Agent · ReAct
- Graph KB lookup
- Multi-turn loop
- extract_slots()

03
CONSULT

- 3 input modes
- Whisper STT
- Pre-visit brief shown
- Adaptive rules injected

04
STRUCTURE

- Report Agent · SOAP
- ICD-10 extract
- Drug interaction (graph)
- update_soap_draft

05
REVIEW

- Evaluator runs · 6 validators / CRITICAL = soft-block / Doctor edits with reason / Edit diff feeds Hermes

06
FINALIZE

- Bilingual EN+MS summary
- Same DB transaction
- VisitFinalized event
- Patient portal push

Next.js 14	Spring Boot 3.3	FastAPI	LangGraph	PostgreSQL	Neo4j 5.20	ILMU GLM 5.1	Whisper-1
APP ROUTER · TS	JAVA 21 · DDD	PYTHON 3.12	REACT LOOP	SUPABASE · 3NF	GRAPH KB	LLM CORE · PROD	STT · OPENAI

Live Demo

PATIENT GRAPH · DRUG GRAPH · MULTI-HOP REASONING

Why a graph, not just a database — multi-hop queries our agents need.

GRAPH · 01

Patient Knowledge Graph

Nodes: Patient, Visit, Symptom, Diagnosis, Medication, Allergy, Condition. Edges link visits → diagnoses & meds. One Cypher hop replaces five SQL JOINs.

▸ Used by Pre-Visit · get_patient_context · get_visit_history (read-only)

GRAPH · 02

Drug Knowledge Graph

Nodes: Drug, DrugClass, Condition, PregnancyCategory. Edges: INTERACTS_WITH (severity), CONTRAINDICATED_IN, UNSAFE_IN, BELONGS_TO.

▸ Used by Evaluator · DDI check · allergy-class cross-check · pregnancy contraindication

GRAPH · 03

Cross-Graph Query

(Patient)-[:ALLERGIC_TO]→(Allergy)←[:BELONGS_TO]-(DrugClass)←[:BELONGS_TO]-(NewDrug). One traversal catches beta-lactam cross-reactivity.

▸ Confidence-scored edges: LLM-extracted vs human-confirmed · queries filter by threshold

GRAPH · 04

Operational Hardening

Schema applied idempotently on FastAPI startup. Bolt protocol for Spring Boot reads. Drainer pattern keeps the graph eventually consistent with Postgres on VisitFinalized.

▸ Idempotent constraints (IF NOT EXISTS) · drainer pattern · audit-grade writes

~8,000 SME CLINICS IN MALAYSIA · PILOT-VALIDATED ECONOMICS

Hospitals get EHRs. Clinics get nothing. We built for the gap.

TAM ~8,000 GP & specialist clinics in Malaysia. SAM ~5,000 SME clinics on paper or basic-digital today. SOM (Year 3) ~300 clinics at conservative 6% penetration. ASEAN expansion (ID/VN/PH) triples the addressable market.

✓ MARKET · MALAYSIA

Pilot-validated unit economics

- **M1** TAM — ~8,000 GP & specialist clinics nationwide
- **M2** SAM — ~5,000 SME clinics on paper or basic digital
- **M3** SOM Y3 — ~300 clinics · 6% penetration · conservative
- **M4** ARPU — RM /clinic/month · tiered by visit volume
- **M5** CAC — Low · MMA + reseller channel partnerships
- **M6** Payback — < 6 months · setup fee covers initial CAC
- **M7** Pilot — 3–5 clinics · 6 months · 50% time-reduction goal
- **M8** Moat — PDPA-by-design · DB-trigger audit · 4-role RBAC

✓ ASEAN · EXPANSION

Same SME profile · 3× TAM · multi-lingual ready

- **F1** ID — Indonesia · ~25,000 SME klinik · BM-adjacent language
- **F2** VN — Vietnam · ~12,000 phòng khám · same SME shape
- **F3** PH — Philippines · ~15,000 clinics · English-friendly
- **F4** Channel — MMA + clinic-mgmt resellers · pre-built networks
- **F5** Partner — MOH · Pharmacy Board · medical training schools
- **F6** Compliance — PDPA shipped · MOH conversation easier
- **F7** Scale — Cloud SaaS · 1-day deploy · no on-prem hardware
- **F8** North star — EHR-grade certification · 3-year horizon

PILOT → NATIONAL → ASEAN · 3-PHASE GTM

PDPA-by-design moat · MMA + reseller channels · MOH conversation underway.

▫ PHASE 1 · KLANG VALLEY PILOT

Q3–Q4 2026

3–5 SME clinics · supervised real-world use

Validate ≥50% documentation-time reduction · zero CRITICAL bypasses

PHASE 2 · NATIONAL

Q1–Q2 2027 · BM intake live

Clinic-management software integrations · resellers

PHASE 3 · ASEAN

Q3 2027+ · ID · VN · PH

Toward EHR-grade certification · e-prescribing

▫ KEY PARTNERS

STRATEGIC + CHANNEL · 3 PILLARS

01	MMA · Malaysian Medical Association	channel	DOCTOR REACH
02	Pharmacy Board · MY	compliance	E-RX READINESS
03	Clinic-Mgmt Resellers	channel	BUNDLED OFFER

▫ COMPLIANCE MOAT · PDPA-BY-DESIGN

Append-only audit log · DB-trigger enforced (never bypassable in app code) · server-side identity from JWT principal · 4-role RBAC · TLS via Nginx · encrypted at rest. MOH conversation easier than competitors.

FIVE ADVANTAGES NO COMPETITOR COMBINES

Agentic. Graph-aware. *Doctor-safe.*

AI scribes only transcribe. Hospital EMRs are too heavy. CliniFlow is the only system built end-to-end as an agentic workflow with a graph KB, around a pluggable LLM, for Malaysian SME clinics.

EDGE · 01

True Agentic Workflow

3 ReAct agents (Pre-Visit · Report · Evaluator) · stateful · per-agent tool allowlists. Not a chatbot bolted onto a form.

EDGE · 02

Clinical-Safety Evaluator

6 validators (DDI, allergy, dose, pregnancy, hallucination, completeness) · 3 enforcement points · doctor override-with-reason.

EDGE · 03

Graph Knowledge Base

Neo4j multi-hop · drug-interaction check · patient memory across visits.

EDGE · 04

Adaptive Rule Learning

Doctor edits → graph rules → injected into future prompts. The system learns.

EDGE · 05

Doctor-Safe by Design

HERMES_FENCE · doctor finalize required · append-only PDPA audit log.

FROM PRELIMINARY MVP TO VALIDATED LAUNCH

Build first. *Test. Then launch.*

We don't ship to clinics until the pilot validates the core hypothesis: ≥50% reduction in documentation time, with no clinically dangerous outputs. Each phase has a clear gate.

PHASE A · BUILD

VHACK 2026 → Q3 2026

Harden the MVP

- Finish all P0 features · 2 agents · 3 input modes
- Drug-interaction graph · adaptive rules
- PDPA hardening · encryption · RBAC · audit
- Internal QA suite · unit + integration tests
- **Gate:** All P0 acceptance criteria pass

PHASE B · TEST

Q4 2026 → Q2 2027

Klang Valley Pilot

- 3–5 SME clinics · supervised real-world use
- Canary → Beta → Majority rollout pattern
- A/B test adaptive rules vs baseline
- Doctor NPS > 30 · edit-rate ≥20% reduction
- **Gate:** ≥50% documentation-time reduction validated

PHASE C · LAUNCH

Q3 2027 →

Public Launch & Scale

- National Malaysia rollout · BM full release
- Clinic-management-software integrations
- ASEAN expansion — ID · VN · PH
- Toward EHR-grade certification · e-prescribing
- **Gate:** Pilot retention & clinical safety metrics held

◉ DEPENDENCY FREE · TANG SHAO XIAN

CLINI FLOW.

Doctors with patients, not paperwork. An agentic workflow built around the LLM as central reasoning core — exactly the way Domain 01 demands.

THANK YOU · QUESTIONS WELCOME